

1. How many accidents occurred in NYC, Austin and Chicago

DAMG7370.FINALPROJECT Settings Code Versions

```
29 WHERE l.City IN ('NewYork', 'Austin', 'Chicago')
30 GROUP BY l.City;
31 select * from accident_fact where accident_id like 'NW%'
32
33 -----
34 1 SELECT
35     l.City AS City,
36     COUNT(DISTINCT a.CRASH_ID_AK) AS cnt_acc
37 FROM DAMG7370.FINALPROJECT.DIM_LOCATION l
38 LEFT JOIN DAMG7370.FINALPROJECT.ACCIDENT_FACT a
39 ON l.Location_ID_SK = a.Location_ID_SK
40 WHERE l.City IN ('NewYork', 'Austin', 'CHICAGO')
41 GROUP BY l.City
42 ORDER BY cnt_acc DESC;
43 -----
44 --2
45 WITH RankedAccidents AS (
```

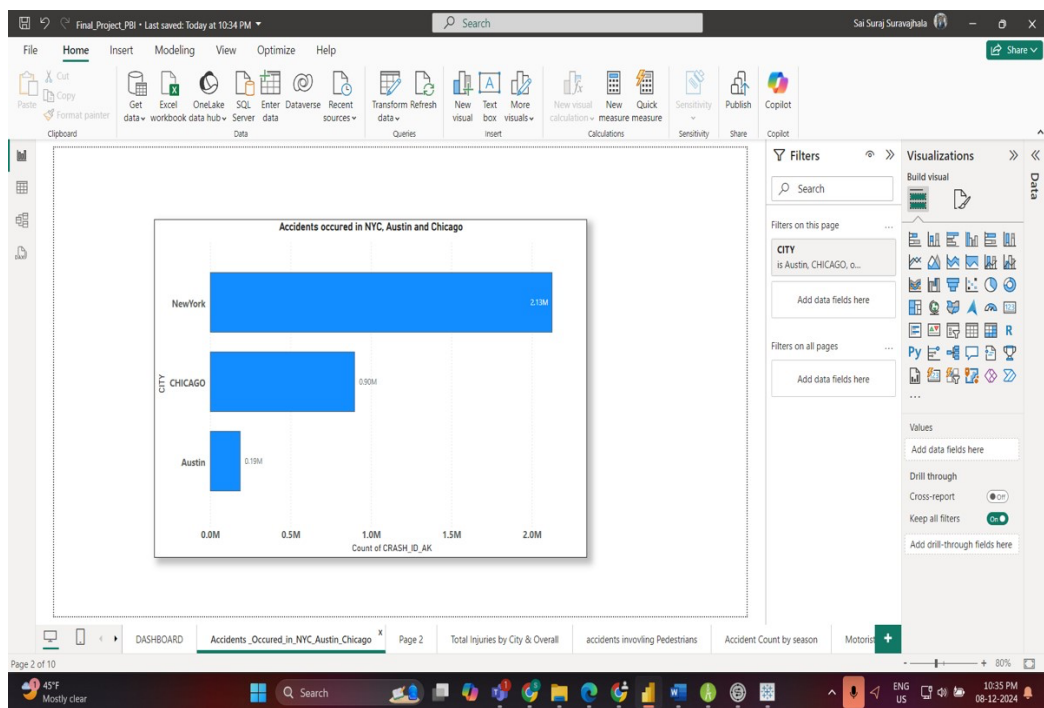
Results Chart

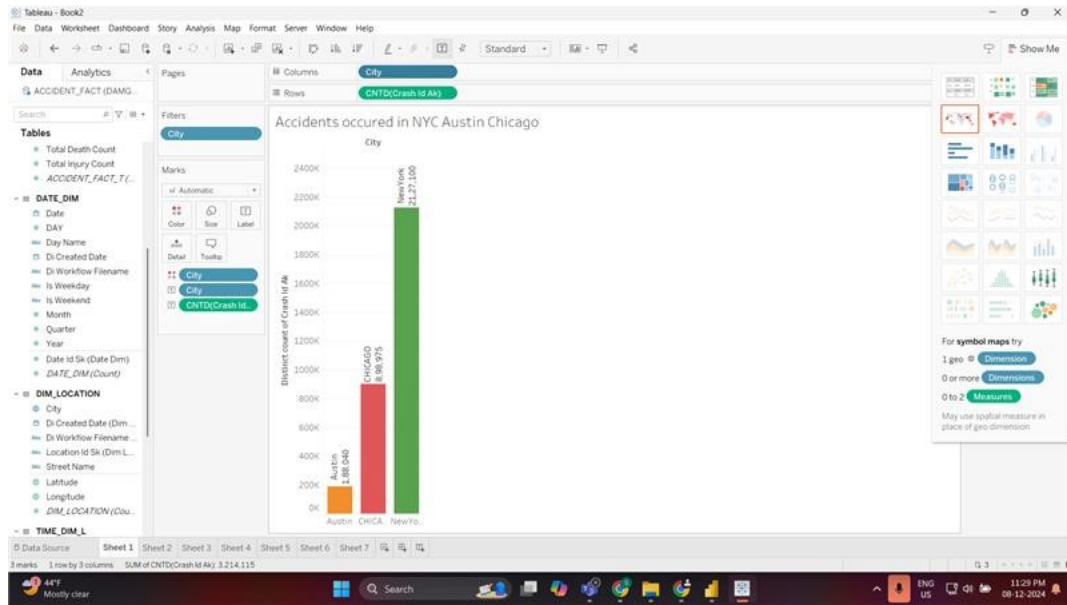
CITY	CNT_ACC
1 NewYork	2127100
2 CHICAGO	899975
3 Austin	188040

Query Details
Query duration 102ms
Rows 3
Query ID 01b8e723-0000-a2e8-0-...

Show more

Ask Copilot





2 Which areas in the cities had the greatest number of accidents

ACCOUNTADMIN • COMPUTE_WH (X-Small) Share

```

--2
WITH RankedAccidents AS (
  SELECT
    l.City,
    l.Street_Name AS Area,
    COUNT(distinct a.crash_id_ak) AS Total_Accidents,
    RANK() OVER (PARTITION BY l.City ORDER BY COUNT(a.crash_id_ak) DESC) AS Rank
  FROM
    DAMG7370.FINALPROJECT.ACCIDENT_FACT a
)

```

Results Chart

CITY	AREA	TOTAL_ACCIDENTS
1 CHICAGO	WESTERN AVE	24493
2 CHICAGO	PULASKI RD	21685
3 CHICAGO	CICERO AVE	20175
4 NewYork	BROADWAY	17277
5 NewYork	ATLANTIC AVENUE	15322
6 Austin	NOT REPORTED	13831
7 NewYork	BELT PARKWAY	13543
8 Austin	LAMAR	5101
9 Austin	IH 35	4197
10 Montgomery	GEORGIA AVE	1203

Query Details

Query duration: 35ms

Rows: 12

Query ID: 01b8e724-0090-a181-0...

Show more

CITY

CHICAGO 3

NewYork 3

Austin 3

Ask Copilot

Final_Project_PBI - Last saved: Today at 10:34 PM

Search

Sai Suraj Suravajhala

File Home Insert Modeling View Optimize Help

Clipboard: Paste, Copy, Format painter

Data: Get data, Excel, OneLake, SQL Server, Data, Transform, Refresh, New visual, Text box, More visuals, New visual calculation, New measure, Quick measure, Sensitivity, Publish, Copilot

CITY	STREET_NAME	Accident_Count
NewYork	BROADWAY	20372
NewYork	NA	456935
NewYork	WESTERN AVE	1
CHICAGO	BROADWAY	4669
CHICAGO	NA	1538
CHICAGO	WESTERN AVE	24493
Austin	BROADWAY	1
Austin	NA	1

Filters: Search, Filters on this page, Add data fields here, Filters on all pages, Add data fields here

Visualizations: Build visual, Add data fields here, Values, Add data fields here, Drill through, Cross-report, Keep all filters, Add drill-through fields here

DASHBOARD Accidents_Occurred_in_NYC_Austin_Chicago Page 2 Total Injuries by City & Overall accidents involving Pedestrians Accident Count by season Motorists

Page 3 of 10

45°F Mostly clear

Search

ENG US 10:36 PM 08-12-2024

3 How many accidents resulted in just injuries?

DAMG7370.FINALPROJECT Settings

Code Versions

```

76 --3 Overall
77
78 SELECT
79     COUNT(DISTINCT a.CRASH_ID_AK) AS total_accidents
80 FROM DAMG7370.FINALPROJECT.ACCIDENT_FACT a
81 WHERE a.Total_Injury_Count > 0 and total_death_count= 0;
82 --BY City
83 SELECT
84     l.City AS city,
85     COUNT(DISTINCT a.CRASH_ID_AK) AS total_accidents
86 FROM DAMG7370.FINALPROJECT.ACCIDENT_FACT a

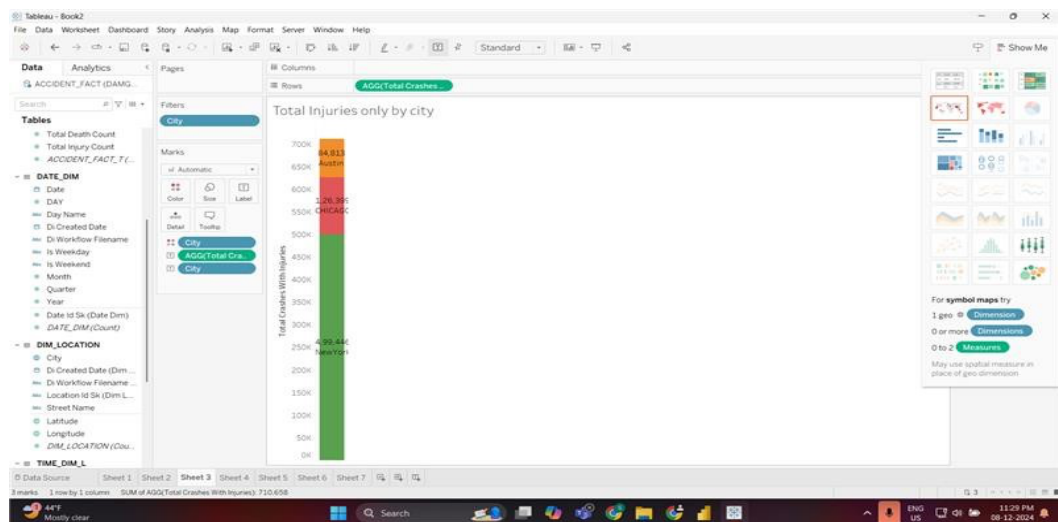
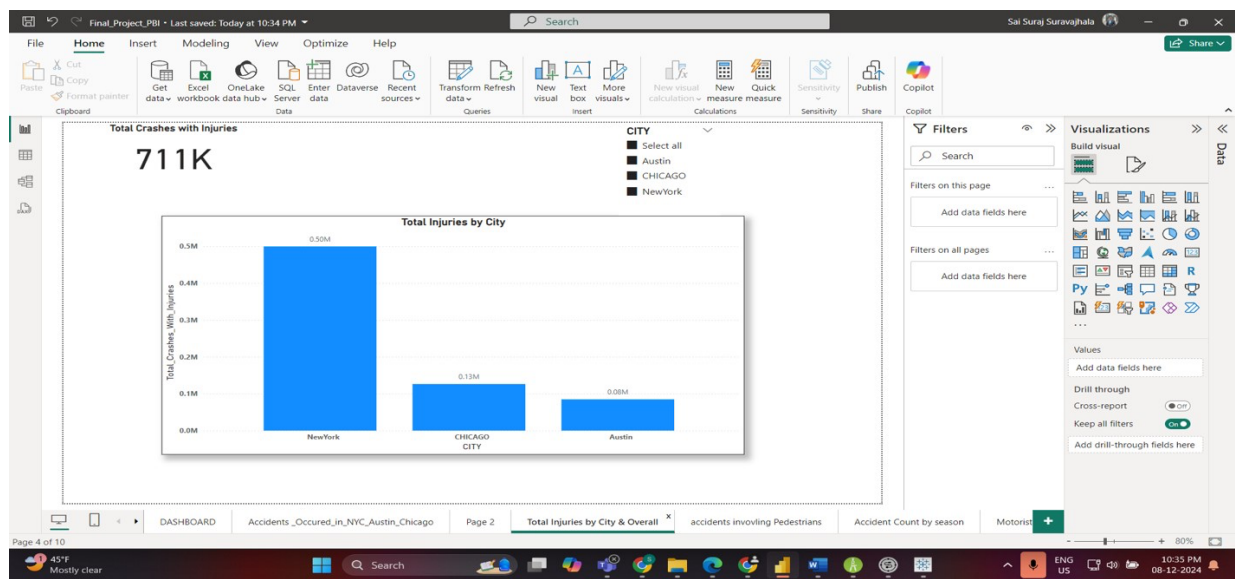
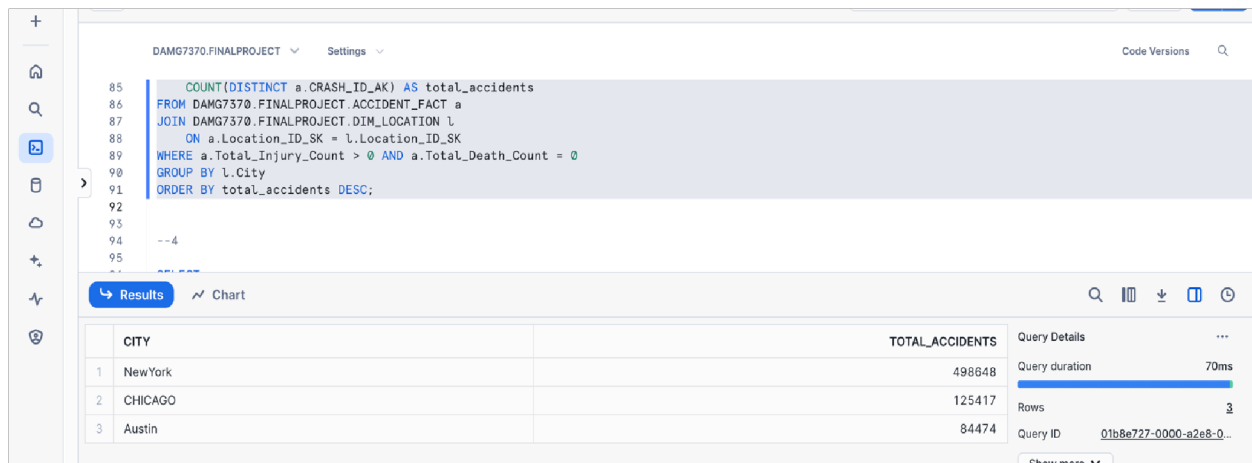
```

Results Chart

	TOTAL_ACCIDENTS	Query Details
1	710864	Query duration 32ms

Rows 1

Query ID 01b8e727-0000-a2e8-0-



4 How often are pedestrians involved in accidents?

The screenshot shows a SQL query editor with the following query:

```
--4
SELECT
  l.city AS city,
  COUNT(DISTINCT fa.crash_id_ak) AS pedestrian_involved
FROM
  accident_fact fa
JOIN
  dim_location l
ON
  fa.location_id_sk = l.location_id_sk
```

The query results are displayed in a table:

CITY	PEDESTRIAN_INVOLVED
1 CHICAGO	21312

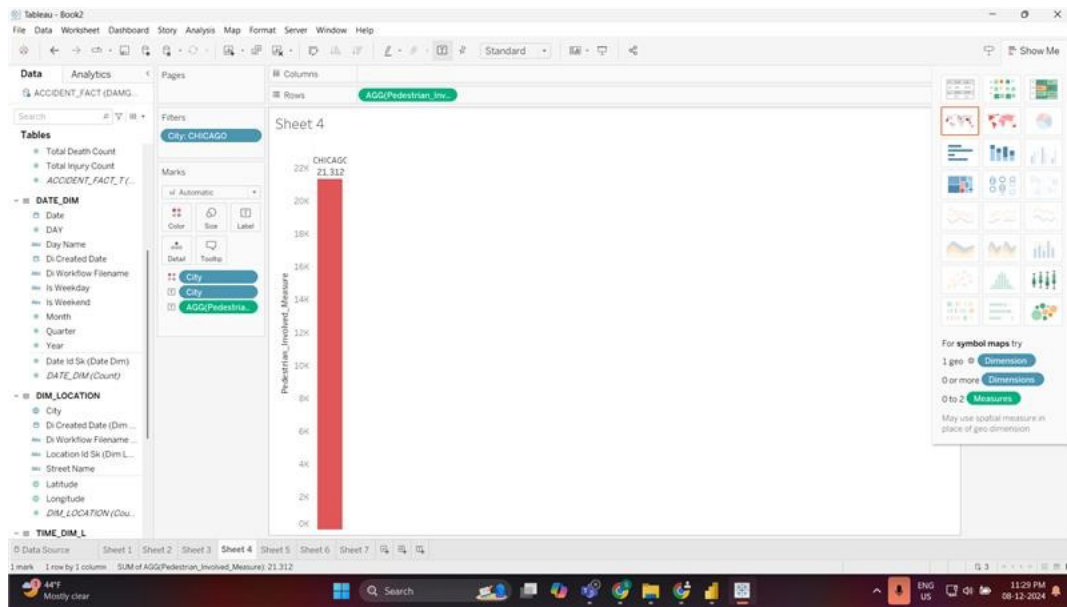
Query Details:

- Query duration: 44ms
- Rows: 1
- Query ID: 01b8e728-0000-a1bd-9...

The screenshot shows a Power BI dashboard with a bar chart titled "Pedestrian_Involved_measure by CITY". The chart displays a single bar for CHICAGO with a value of 21.3K. The Y-axis is labeled "Pedestrian_Involved_measure" and ranges from 0K to 20K. The X-axis is labeled "CITY".

The dashboard includes a top navigation bar with tabs: DASHBOARD, Accidents_Occurred_in_NYC_Austin_Chicago, Page 2, Total injuries by City & Overall, accidents involving Pedestrians (selected), Accident Count by season, and Motorists.

The bottom status bar shows the date and time: 10:35 PM 08-12-2024.



5 When do most accidents happen (In which season)?

```

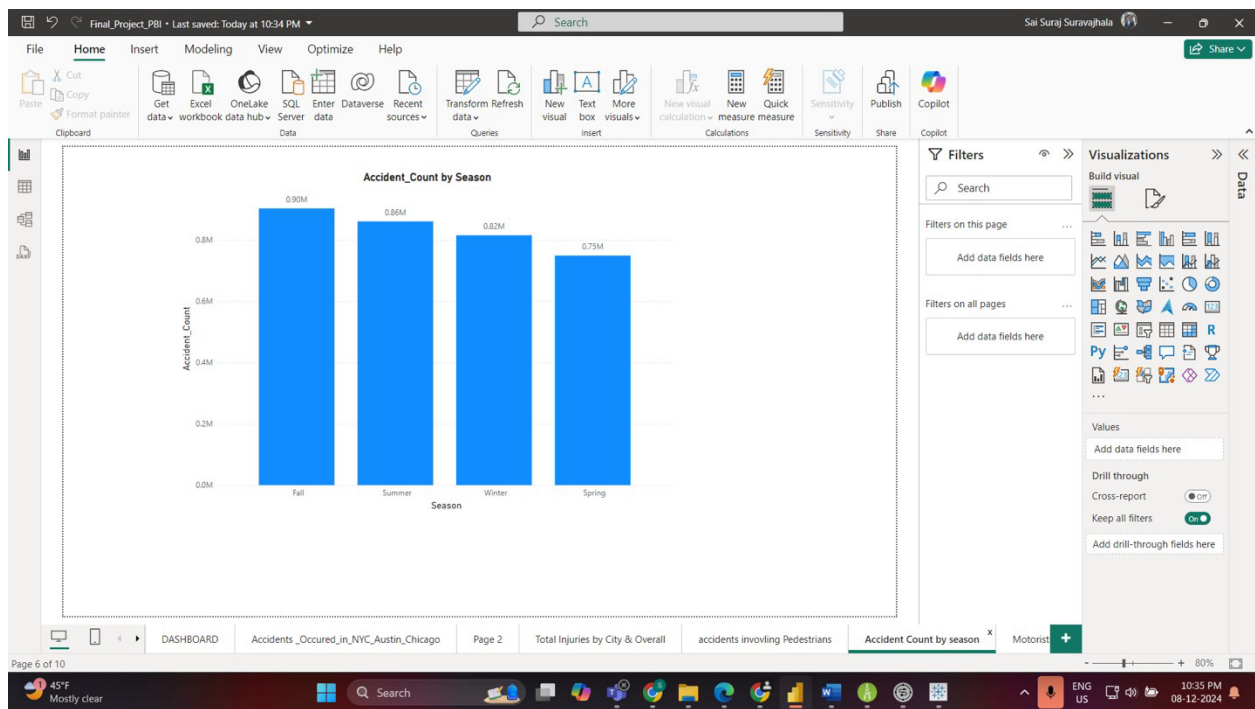
111 SELECT SUM(Accident_Count)
112
113 ---
114
115
116 SELECT
117 CASE
118   WHEN d.month IN (11, 12, 1) THEN 'Winter'
119   WHEN d.month IN (3, 4, 2) THEN 'Spring'
120   WHEN d.month IN (6, 7, 5) THEN 'Summer'
121   WHEN d.month IN (9, 10, 8) THEN 'Fall'
122   ELSE 'Unknown'

```

SEASON	ACCIDENT_COUNT	# ACCIDENT_COUNT
1 Fall	903799	749631
2 Summer	861277	
3 Winter	815844	
4 Spring	749631	

AK

Ask Copilot



6 How many motorists are injured or killed in accidents?

ACCOUNTADMIN COMPUTE_WH (X-Small) Share

```

169 --6
170
171 SELECT
172     SUM(a.Motorist_Injured) AS Motorist_Injured,
173     SUM(a.Motorist_Killed) AS Motorist_Killed
174 FROM
175     DAMG7370.FINALPROJECT.ACCIDENT_FACT a;
176
177
178 SELECT
179     l.City AS City,
180     SUM(a.Motorist_Injured) AS Motorist_Injured,
181     SUM(a.Motorist_Killed) AS Motorist_Killed
182 FROM
183     DAMG7370.FINALPROJECT.ACCIDENT_FACT a
184 JOIN
185     DAMG7370.FINALPROJECT.DIM_LOCATION l
186 ON
187     a.Location_ID_SK = l.Location_ID_SK
188 GROUP BY
  
```

Results Chart

	MOTORIST_INJURED	MOTORIST_KILLED
1	1253899	4170

Query Details: Query duration 151ms, Rows 1, Query ID 01b6, Ask Copilot, Show more

Browser tabs: Gmail, YouTube, Maps, Gmail, YouTube, Maps, News. Address bar: 2024-12-05 6:45pm, 2024-12-05 7:02pm, 2024-12-05 7:06pm, 2024-12-05 7:10pm, 2024-12-05 7:10pm, Load sample data with Pyt..., 2024-12-05 7:40pm. User: ACCOUNTADMIN, COMPUTE_WH (X-Small). Share button.

```
174 DAMG7370.FINALPROJECT
175
176 DAMG7370.FINALPROJECT.ACCIDENT_FACT a;
177
178 SELECT
179     t.CITY AS City,
180     SUM(a.Motorist_Injured) AS Motorist_Injured,
181     SUM(a.Motorist_Killed) AS Motorist_Killed
182 FROM
183     DAMG7370.FINALPROJECT.ACCIDENT_FACT a
184 JOIN
185     DAMG7370.FINALPROJECT.DIM_LOCATION L
```

Results | Chart

CITY	MOTORIST_INJURED	MOTORIST_KILLED
1 NewYork	1242115	3460
2 Austin	5275	676
3 CHICAGO	666	0
4 Montgomery	0	0

Query Details: Query duration 428ms, Rows 4, Query ID 07b8e72c-0000-a0ef-0...

Ask Copilot

Final_Project_PBI - Last saved: Today at 10:34 PM. Search bar. User: Sai Suraj Suravajhala. Share button.

Home | Insert | Modeling | View | Optimize | Help

Clipboard: Cut, Copy, Paste, Format painter. Data sources: Get, Excel, OneLake, SQL Server, Enter data, Database, Recent sources. Queries: Transform data, Refresh. Visuals: New visual, Text box, More visuals, Insert. Calculations: New visual calculation, New measure, Quick measure, Sensitivity, Sensitivity, Publish, Copilot.

Motorists Killed and Injured by City

Sum of MOTORIST_KILLED (blue), Sum of MOTORIST_INJURED (light blue)

CITY	Sum of MOTORIST_KILLED	Sum of MOTORIST_INJURED
NewYork	3460	1242115
Austin	676	5275
CHICAGO	0	666
Montgomery	0	0

Motorists Killed

4.17K (0.00K to 8.34K)

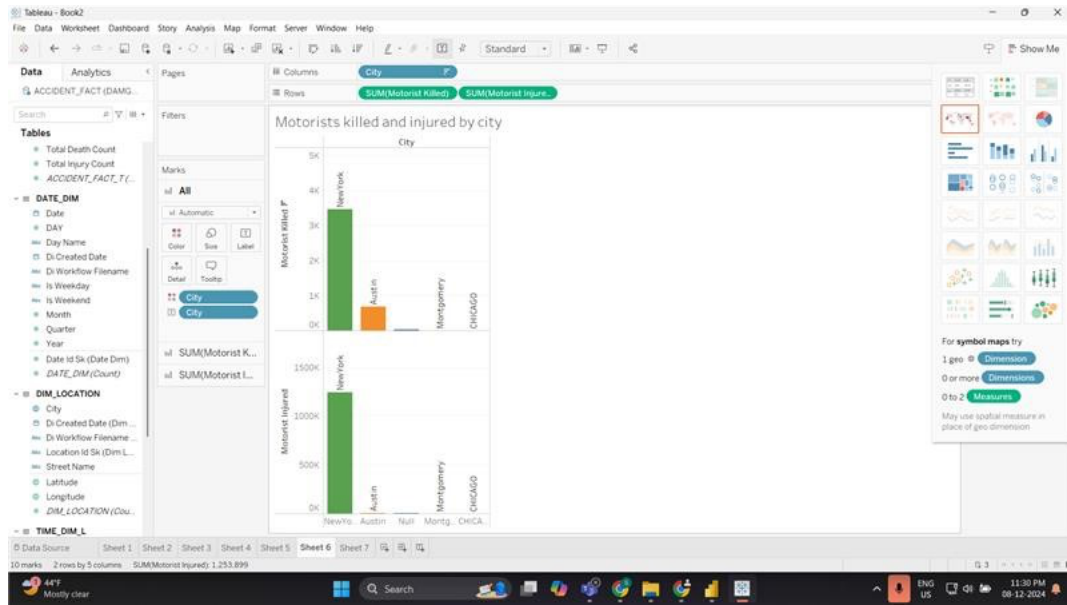
Motorists Injured

1.25M (0.00M to 2.51M)

Filters: Search, Add data fields here. Visualizations: Build visual, Add data fields here. Values: Add data fields here. Drill through: Cross-report (OFF), Keep all filters (ON), Add drill-through fields here.

Page 7 of 10. City & Overall | accidents involving Pedestrians | Accident Count by season | Motorists Killed and Injured by City | Time based Analysis of accidents | 7th | Fatality Analysis

45°F Mostly clear. Search. 10:36 PM 08-12-2024.



7 Which top 5 areas in the cities have the most fatal number of accidents?

5:04:5pm 2024-12-05 7:02pm 2024-12-05 7:06pm 2024-12-05 7:10pm Load sample data with Pyt... 2024-12-05 1:40pm ACCOUNTADMIN COMPUTE_WH (X-Small) Share

DAMG7370.FINALPROJECT Settings Code Versions

```

196 -----
197 --7
198
199 WITH RankedFatalAccidents AS (
200     SELECT
201         1.City,
202         1.Street_Name AS Area,
203         COUNT(a.Accident_ID) AS Fatal_Accidents,
204         RANK() OVER (PARTITION BY 1.City ORDER BY COUNT(a.Accident_ID) DESC) AS Rank
205     FROM
206         DAMG7370.FINALPROJECT.ACCIDENT_FACT a

```

Results Chart

CITY	AREA	FATAL_ACCIDENTS
1 Austin	NOT REPORTED	90
2 NewYork	BROADWAY	74
3 NewYork	3 AVENUE	73
4 NewYork	BELT PARKWAY	65
5 NewYork	ATLANTIC AVENUE	60
6 NewYork	LINDEN BOULEVARD	57
7 CHICAGO	PULASKI RD	32
8 Austin	LAMAR	31
9 CHICAGO	HALSTED ST	31

Query Details

Query duration 361ms

Rows 15

Query ID 01b8e72e-0000-a1bd-0...

Show more

CITY

Austin

NewYork

CHICAGO

8 Time-based analysis of accidents

- Time of the day, day of the week, weekdays, or weekends

```
241 DAMG7370.FINALPROJECT.ACCIDENT_FACT a
242 JOIN
243 DAMG7370.FINALPROJECT.TIME_DIM t
244 ON
245 a.TIME_ID_SK = t.TIME_ID_SK
246 GROUP BY
247 t.Time_Period;
248
249 --Count By day
250
251
252 SELECT
```

TIME_PERIOD	COUNT_ACC
1 Morning	1690421
2 Afternoon	273422
3 Late Night	261317
4 Early Morning	1105391

Query Details: Query duration 678ms, Rows 4, Query ID 01b8e72d-0000-a0b6...

TIME_PERIOD: 100% filled

COUNT_ACC

Ask Copilot

```
254 COUNT(DISTINCT a.CRASH_ID_SK) AS Count_Acc
255 FROM
256 DAMG7370.FINALPROJECT.ACCIDENT_FACT a
257 JOIN
258 DAMG7370.FINALPROJECT.DATE_DIM d
259 ON
260 a.Date_ID_SK = d.Date_ID_SK
261 GROUP BY
262 d.day_name;
263
264
265
```

DAY	COUNT_ACC
1 Thu	492769
2 Wed	482730
3 Sun	404461
4 Mon	469151
5 Sat	461849
6 Tue	485704
7 Fri	533887

Query Details: Query duration 707ms, Rows 7, Query ID 01b8e72d-0000-a334...

DAY: 100% filled

COUNT_ACC

Ask Copilot

+

Home

Search

Visuals

Modeling

View

Optimize

Help

DAMG7370.FINALPROJECT

Settings

Code Versions

270

271

272

273

274

275

276

277

278

279

280

FROM

DAMG7370.FINALPROJECT.ACCIDENT_FACT a

JOIN

DAMG7370.FINALPROJECT.DATE_DIM d

ON

a.Date_ID_SK = d.Date_ID_SK

GROUP BY

d.Is_WeekDay;

Results

Chart

IS_WEEKDAY	COUNT_ACC
1 N	866310
2 Y	2464241

Query Details

Query duration 645ms

Rows 2

Query ID 01b8e72d-0000-a181-0...

Show more

IS_WEEKDAY 100% filled

COUNT_ACC

Ask Copilot

AK

1

2

3

Final_Project_PBI - Last saved: Today at 10:34 PM

Search

Sai Suraj Suravajhala

Share

File

Home

Insert

Modeling

View

Optimize

Help

Cut

Copy

Format painter

Clipboard

Get data

Excel workbook

OneLake data hub

SQL Server Data

Enter data

Dataaverse

Recent sources

Transform data

Refresh data

Queries

New visual

Text box

More visuals

New visual calculation

New measure

Quick measure

Sensitivity

Publish

Copilot

No. of Accidents by Day

No. of Accidents by Weekday & Weekends

IS WEEKDAY

Time based Analysis of accidents

Filters

Visualizations

Data

City & Overall

accidents involving Pedestrians

Accident Count by season

Motorists Killed and Injured by City

Time based Analysis of accidents

7th

Fatality Analysis

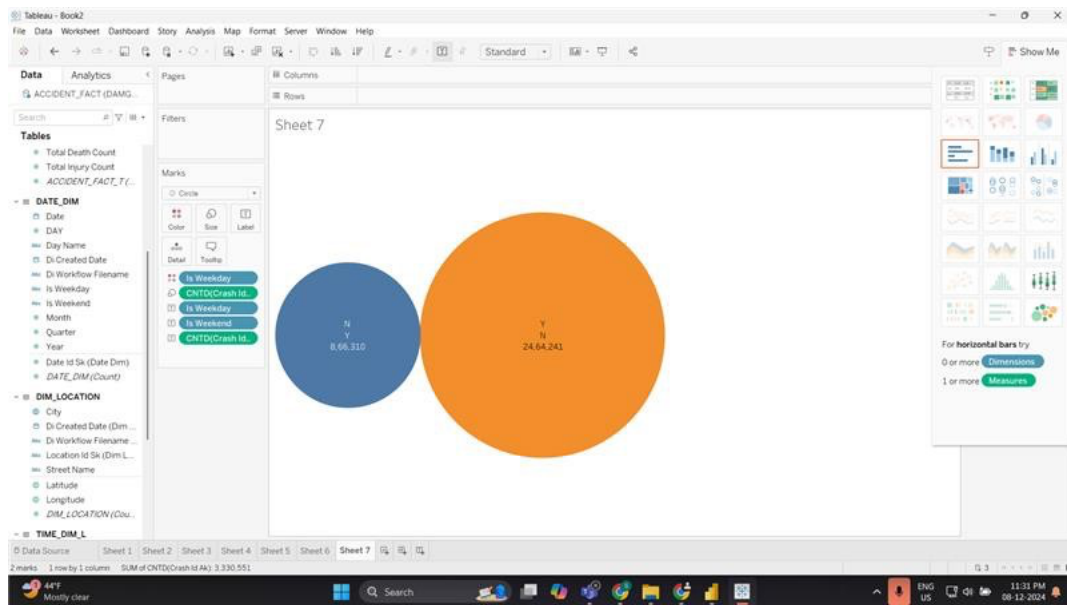
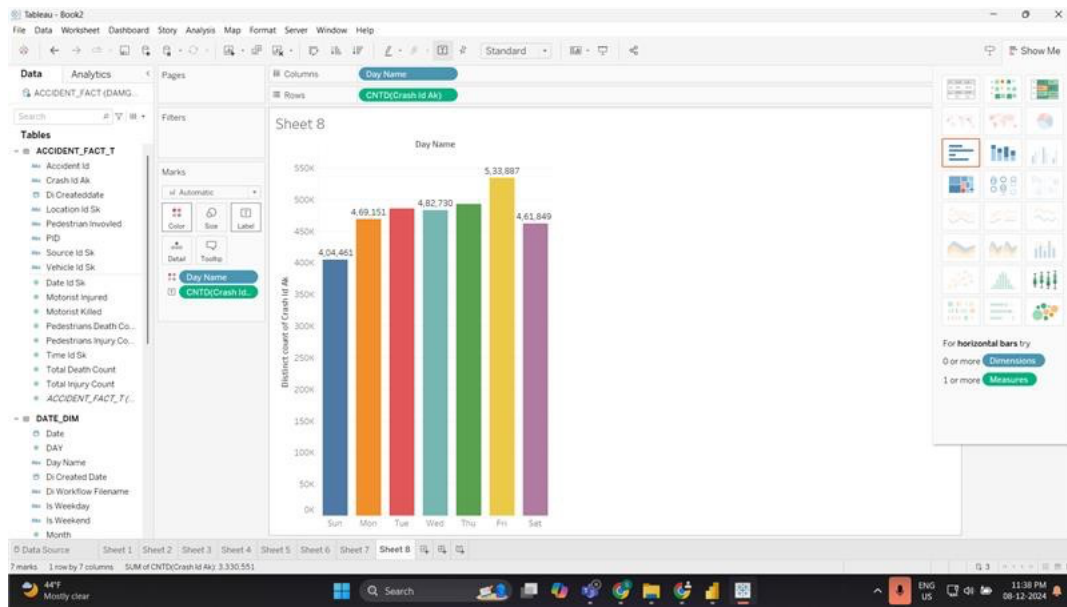
Page 8 of 10

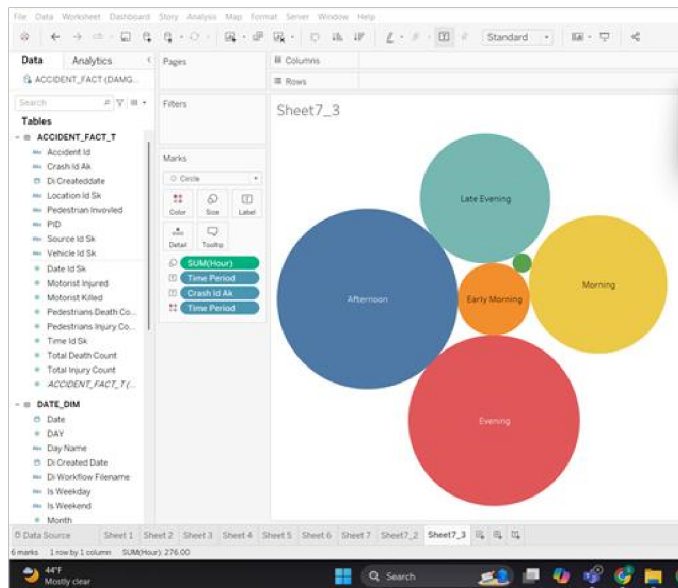
43°F Mostly clear

Search

ENG US

10:36 PM 06-12-2024

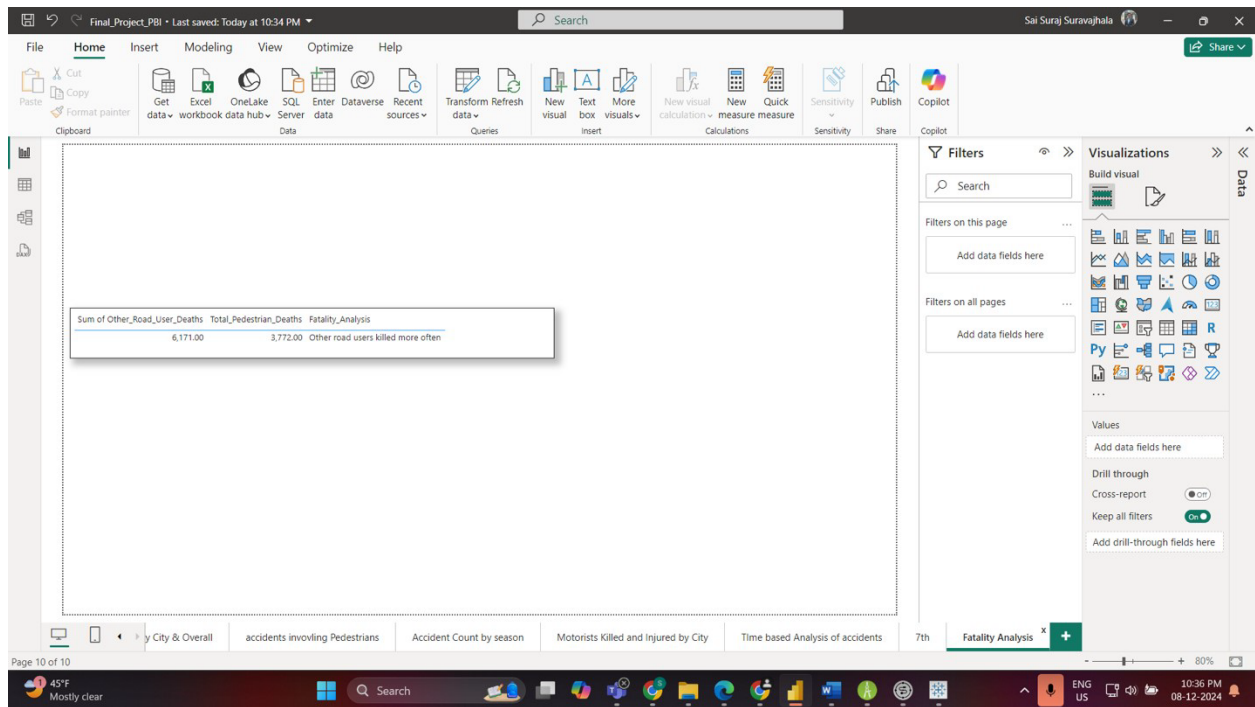




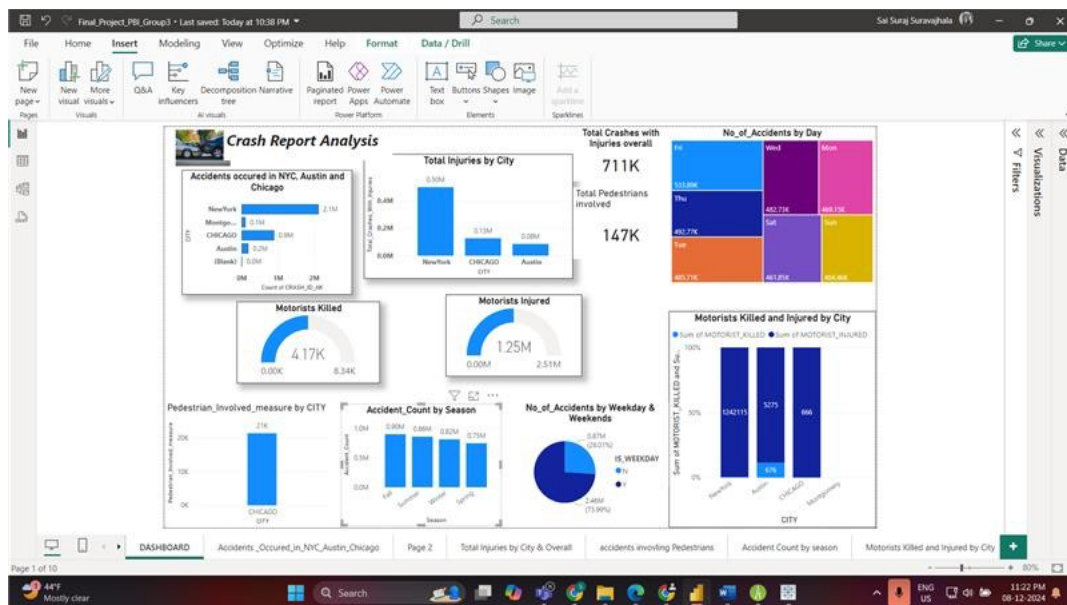
9 Fatality analysis

- Are pedestrians killed more often than road users?





Dashboard of Power BI



Location based Analysis with respect to Count of Crash Id

